

CURRICULUM VITAE

Dr. MUVVALA GOPI NATH

Assistant Professor

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PERSONAL DETAILS

Date of Birth	Gender	Category
17-08-1990	Male	OBC

PUBLICATIONS RECORD

Publications	h-index	i10 - Index
43	21	29

ACADEMIC QUALIFICATION

Degree/Exam	University/Board	Year of passing	%marks /CGPA	Branch/Field
PhD	IIT Kharagpur	2018	-	Online Monitoring of Laser based Additive Manufacturing
Master of Technology	IIT Kharagpur	2013	9.54	Manufacturing Science & Engineering
Bachelor of Engineering	Gandhi Institute of Technology and Management (GITAM)	2011	8.82	Mechanical Production and Industrial Engineering

WORK EXPERIENCE

S. No.	Position held	Name of the Institute	Period	
			From	To
1.	Senior Scientific Officer	Indian Institute of Technology Kharagpur	18-05-2018	10-04-2019
2.	Assistant Professor	Indian Institute of Technology Hyderabad	15-04-2019	Till date

AREAS OF INTEREST

- Additive Manufacturing
- Laser Surface Engineering: Cladding, Alloying, Remelting and Transformation hardening
- Laser Welding: Similar and dissimilar materials
- Under Water Laser Material Processing
- Laser Forming
- Solid State Joining and Additive Manufacturing
- Electric Discharge Machining
- Online monitoring of laser material processing

PROFESSIONAL RECOGNITION/ AWARD

S. No	Name of Award	Awarding Agency	Year
1	INAE Young Associate	Indian National Academy of Engineering (INAE)	2024

BEST PAPER / POSTER AWARDS

S. No	Title of paper	Name of the conference	Authors list	Year of Conference
1	Laser surface polishing of additively manufactured components through laser beam shaping	International Conference on Materials Processing using Lasers	Arkajyoti Jha, Vignan Babu Dyavanapelly, Bikram K Khandai, Muvvala Gopinath	2023

		and Surface Engineering (IMPULSE 2023), IIT Madras		
2	A Study on the Effect of Modulated Mode Laser on Anisotropy in Laser-Directed Energy Deposition	International Conference on Materials Processing using Lasers and Surface Engineering (IMPULSE 2023), IIT Madras	Arkajyoti Jha, Dharavathu Raja, Bikram K Khandai, Muvvala Gopinath	2023
3	Strategies to Control Porosity in Laser Directed Energy Deposition	International Conference on Materials Processing using Lasers and Surface Engineering (IMPULSE 2023), IIT Madras	Arkajyoti Jha, Jagdish Chandra Pandey, Bikram K Khandai, Muvvala Gopinath	2023
4	Controlling waviness in additive manufacturing of thin walls by laser directed energy deposition process	8th International and 29th All India Manufacturing Technology (AIMTDR)	Srinath Gudur, Shivam Shukla, John John Rozario Jegaraj, Mastanaiah P, Muvvala Gopinath, Suryakumar Simhambhatla	2021
5	A study on the effect of cooling rate on evolution of microstructure in laser surface remelting of Inconel 718	ICRAMERD	Arkajyoti Jha, Amit Choudhary, Batta Renuka, Muvvala Gopinath	2020

INFRASTRUCTURE ESTABLISHED

S. No	Equipment Name	Funding Source
1	40 kN Friction Stir Welding Machine	JICA
2	1.5 Kw Yb-Fiber Laser with Interchangeable Optics for Multipurpose Laser Material Processing	Project
3	Q-Switched Nd:YAG Laser (1 J Nano second Laser)	Project
4	Green Laser based Powder Bed Fusion System	Project
5	Resistance Seam welding system	Project
6	6-axis Robotic Arm with Weld Table	Project
7	Multi-Wire Feed TIG Welding System	Project
8	Machining Center (Vertical Milling Center)	Project + Pooling funds from MAE Department Colleagues
9	Mutli-Purpose 3-axis Gantry Type CNC with Rotatory Axis	Project
10	Thermal Imaging Camera	Project
11	Infrared pyrometers (4 units)	Project
12	Laser Displacement Sensor	Project
13	Stereolithography & FDM Printers with Curing Machine (3 Units)	Project
14	Workstations (7 Units)	Project
15	Laser Engraver	Project
16	3D Scanner	Project
17	Materialise Magics RP + import + lattice + nester module (Licenses: 15)	Project
18	Simufact Software Package with Additive Manufacturing and Welding Modules	Project
19	Optical and 3D Welding Tables	Project
20	Precision Linear Stages	Project
21	High Temperature Tribometer (Brought Back to operation with fund of 29 Lakhs)	Project + Institute
22	Ultrasonic Vibration Stage	Project
23	Laser Directed Energy Deposition Head	Project
24	Rotary Screw Compressor	Project

LIST OF STUDENTS SUPERVISED

Sl. No.	Name	Project title/Domain	Graduation year	Guide/Co-guide/PI	Current position
Postdoctoral Researcher					
1	Dr. Amit Choudhary	Additive Manufacturing & its Real-time Monitoring	2021	PI	Assistant Professor, IIT Roorkee
PhD Students					
1	Dr. Bikram Keshari Khandai	Developing a fundamental insight into the bending mechanism in a multi-pass laser forming through real-time monitoring	2025 (Completed)	Guide	Postdoctoral researcher at IIT Hyderabad
2	Arkajyoti Jha	Additive Manufacturing & its Real-time Monitoring	Ongoing	Guide	NA
3	Raja Dharavathu	Laser Assisted Hybrid friction Stir welding and Laser Directed Energy deposition	Ongoing	Guide	NA
4	Ganesh Sankar	Laser softening and its real-time monitoring	Ongoing	Guide (External Student)	NA
Master's Students					
1	Shivam Shukla	Real-Time Monitoring of Temperature Gradients and Bending Mechanism in Multi-Scan Laser Forming	2021	Guide	Ph.D. in KU Leuven, Belgium
2	Sushmita Banda	A Study on Laser Powder Interaction in Laser Directed Energy Deposition Process	2021	Guide	PhD in Purdue university, Operations Manager, Amazon
3	Vignesh Karthikeyan	Strategies to Control Heat Accumulation and Anisotropy in Metal Additive Manufacturing Process	2022	Guide	Senior Associate, PwC
4	Pravesh Jhanwar	Ultrasonic Vibration Assisted Additive Manufacturing	2022	Guide	Business Analyst, Nineleaps
5	Nasreldein Daoud Suilman Abdlla	Laser Welding of Inconel 718: Influence of scanning speed on evolution of Microstructure and Mechanical properties	2022	Guide	Manager, Yosir Advanced Business Co. Ltd.
6	Kishan	Machinability Studies on Additively Manufactured Inconel 718 and Additively Manufactured and Heat Treated Titanium Alloy Ti-6Al-4V	2022	Guide	Senior Associate, PwC
7	Dyavanapelly Vignan Babu	A Fundamental Insight into Extreme High-Speed Laser Cladding and Laser Beam Shaping for Surface Engineering Applications	2023	Guide	Applications Engineer, KLA
8	Chilukuri Sneha	Effect of scan strategy and heat treatment on impact toughness of Laser Directed Energy Deposited Inconel 718	2023	Guide	CAE Simulation Engineer, Mercedes-Benz
9	Mothukupally Prashanth	Local Weld Characterization of Laser Welded Inconel 718 sample Using DIC Technique	2023	Co-guide	Senior Engineer, Whirlpool Corporation
10	Rahul Singh	Mechanical Behaviour of Hybrid Components Fabricated using Wire Arc Additive Manufacturing Process	2024	Guide	Senior Executive, TATA Advanced System

11	Jagdish Chandra Pandey	Development of Friction Stir Welding tool of Stellite 6 by L-DED process	2024	Guide	Senior Manager, TATA Motors
12	Yash Tandel	Laser welding of low alloy steel	2024	Guide	Associate Product Manager, Wealthy
13	Sachin Labana	A study on thermal management to control the Residual Stress & mitigate cracks in Thermal Barrier Coating (TBC) deposited by L-DED and Benchmarking its Performance by Plasma Spray Coating	2024	Guide	Officer Engineer, HPCL
14	Dawke Pratik Koyalkar	Additive Manufacturing of Bottom Clamp for Rocker Arm using WAAM Process: Process Optimization	2025	Guide	NA
15	Biradar Prasannakumar Dattatraya	Design and Development of Friction stir welding tool and joining of high strength material using FSW	2025	Guide	Product Application Engineer, KLA
16	Ashish Parihar	A Study on the influence of Shielding Gas on the Microstructure and Mechanical performance of Additively Manufactured Inconel 718 at Room and Elevated Temperature	2025	Guide	Indian Navy
17	Aditya Manoj Madgulkar	Machine Learning and Language Model for Intelligent Control and Knowledge Retrieval in Additive Manufacturing	2025	Guide	PGET, Pratt & Whitney
18	Dhakshayani Duppa	Laser DED of 17-4 PH steel: Effect of heat input and post deposition heat treatment on microstructure and properties	2025	Co-guide	NA

WORKSHOPS/CONFERENCE/ TRAINING PROGRAMS ORGANISED

S. No.	Role	Conference Name	Venue	Date & Year
1	Session Chair	Aerospace Manufacturing and Assembly, AeroCON 2026	SAEINDIA in Bengaluru	4 th & 5 th June 2026
2	Chair	One day Workshop on Welding/Joining Techniques for Vikasith Bharath/Atmanirbhar Bharth, Indian Institute of Welding - India	IIT Hyderabad	23 rd January 2026
3	Technical Coordinator	1st International Conference on Frontiers in Surface Engineering and Additive Manufacturing (FSEAM 2026)	IIT Kharagpur	21 st – 23 rd January 2026
4	Co-chair	7th International and 22nd National Conference on Machines and Mechanisms,	IIT Hyderabad	7 th -10 th December 2025.
5	Chairperson and Session Chair	Core Organizing Committee for IIM Annual Technical Meeting		4 th – 6 th December 2025
6	Organizer	Two days hands on training workshop on additive manufacturing simulation		13 th & 14 th October 2025
7	Resource Person	Two days hands-on workshop on DED Metal Additive Manufacturing – Applications and Challenges	V. R. Siddhartha Engineering College	22 nd & 23 rd August 2025
8	Co-organizer,	Workshop on additive manufacturing for scientists from DRDO labs	IIT Hyderabad	24 th – 28 th February, 2025.
9	Co-Convener	PMRF Annual Symposium 2025		16 th & 17 th March 2025

10	Co-Convener	Processing of Advanced Materials and Fabrication of Products (PAMFP)		11 th – 13 th December, 2024
11	Resource Person	One-day training programme on Additive Manufacturing Technologies	Online across all units of BHEL executives in India	24 th July 2024
12	Resource Person	Two day training program on Metal Additive Manufacturing using DED technology		7 th & 8 th September 2023
13	Resource Person	One day offline training program on Design for Additive Manufacturing	BHEL Corporate R&D, Hyderabad	17 th & 18 th July 2023
14	Organizing Member	4th Structural Integrity Conference and Exhibition	IIT Hyderabad	14 th – 16 th December 2022

INVITED TALKS

S. No	Nature of Talk	Nature of the event	Event Name	Title	Organizing institute	Year
1	Invited Speaker	International Conference	1st International Conference on Frontiers in Surface Engineering and Additive Manufacturing (FSEAM 2026)	Extreme high speed laser cladding of highly reflective and highly conductive materials	IIT Kharagpur	Jan 2026
2	Resource speaker	Workshop	Metal Additive Manufacturing: Recent Progress and Future Opportunities	Strategies to control porosity in laser DED	NIT Rourkela	Jul 2025
3	Keynote Speaker	National Symposium	National symposium on Advancements in Metal Additive Manufacturing	Latest Advancements and Innovations in Wire Arc Additive Manufacturing	Vikram Sarabhai Space Centre (VSSC)	May 2025
4	Speaker	Workshop	Powder Characterization Workshop	Optimizing Additive Manufacturing: The Impact of Powder Morphology and Characteristics	IIT Hyderabad	Mar 2025
5	Invited talk	International Conference	International Conference on Powder Metallurgy & Particulate Materials	Exploring the Role of Laser-Powder Interaction in Controlling Inter-Layer Porosity during Laser-Directed Energy Deposition	Powder Metallurgy Association of India	Feb 2025
6	Invited talk	International Conference	International conference on Laser & other deposition techniques	Role of laser material interaction in the deposition of thin coatings using extreme high-speed laser cladding	IIT Hyderabad & IIT Madras	Mar 2025
7	Invited talk	International Conference	International Conference on Additive Manufacturing and Characterization (ICAMC) Conference	The Impact of Real-Time Monitoring on Geometrical, Microstructural, and Quality Control in Additive Manufacturing	IIT Bombay	Jan 2025
8	Resource speaker	Atal FDP	Advances in Manufacturing: Special Processes and Practices	Laser Surface Engineering and its real- time monitoring	BIT Sindri	Dec 2024
9	Invited talk	Karyashala	Metal additive manufacturing fundamentals and applications	Process monitoring in Additive Manufacturing	IIT Delhi	Jan 2024

10	Invited Talk	Training Program	Training Program	Introduction to Additive Manufacturing	Military College of Electronics & Mechanical Engineering (MCME)	Oct 2024
11	Invited talk	Karyashala	Novel Materials: Processing and Characterization	Process Control in Additive Manufacturing	SVNIT, Surat	Mar 2024
12	Resource Person	Additive Technology Enhancement Program	Additive Manufacturing: Techniques, Materials, Applications and Future Trends	Insights into Laser Directed Energy Deposition	Vel Tech	Dec 2023
13	Invited speaker	Short term course	Laser Based Manufacturing for Future Industrial Development	Advances in Laser Based Additive Manufacturing	NIT Rourkela	Jul 2023
14	Invited speaker	Workshop	Research Trends in Additive Manufacturing Technologies (RTAM)	Metal additive Manufacturing	Prayojaka Technologies Pvt Ltd	Aug 2023
15	Expert Talk	Workshop	Monitoring and Characterization of Metal Additive Manufacturing, M&C@MAM-2022	Role of Monitoring in laser material processing	VNIT Nagpur	Dec 2022
16	Invited Talk	International Conference	International Conference on Laser Assisted Materials Processing	Laser Directed Energy Deposition of Inconel 718 and Its mechanical Characterization	IIT Kharagpur	Aug 2022
17	Resource person	ATAL Workshop	Additive Manufacturing: Lab Scale to Field Scale	real time monitoring and control in additive manufacturing	IIT Palakkad	Dec 2021
18	Resource person	STTP	Rapid Prototyping and Digital Manufacturing	Anisotropy in Additive Manufacturing	NSS College of Engineering Palakkad	Nov 2020
19	Speaker	Pre-workshop conference	4th Structural Integrity Conference and Exhibition	Additive Manufacturing	IIT Hyderabad	Dec 2022
20	Invited talk	Karyashala	“Advances in Welding and Additive Manufacturing (AWAM)”	Fundamentals of Laser Welding	SVNIT, Surat	Jul 2022
21	Resource person	AICTE-QIP	Micro and Precision Manufacturing	Real-time monitoring of molten pool thermal history in Additive Manufacturing	IIT Indore	Oct 202
22	Resource person	TEQIP III	Advances in Surface Engineering: Special Process & Practices	Laser surface engineering and its real-time monitoring	BIT Sindri	Dec 2020
23	Resource person	FDP	Modeling, Simulation and Experimental Approaches of Mechanical Systems	Thermal Management in Wire Arc Additive Manufacturing	GCEK, Bhawanipatna	Aug 2020

ADMINISTRATIVE ACTIVITIES AT IIT HYDERABAD

S. No.	Role	Section	Institute/Department Level	Duration
1	Chair	Hostel Coordinating Unit	Institute	July 2024 – Till date
2	Coordinator	Joint Centre of Excellence on Advanced Manufacturing Research, Innovation and Training (AMRIT), with Deakin University, Australia	Institute	2025
3	Program Coordinator	MTech in Lightweighting Engineering	Department	2025
4	Core Committee Member	Inter IIT Sports meet 2025	Institute	2025
5	Faculty Placement coordinator	MTech in Additive Manufacturing	Department	2023 - 2025
6	Associate Chair	Hostel Coordinating Unit	Institute	June 2023 – July 2024
7	Admissions coordinator	MTech (COAP) in Additive manufacturing	Department	2020 - 2024
8	Coordinator	Coordinator for establishment of amenities in International Guest House at IIT Hyderabad	Institute	2023
9	Admission Coordinator	MTech in Department of Mechanical and Aerospace Engineering	Department	2020 - 2021
10	Warden	Hostel Coordinating Unit	Institute	2019 – 2022

PEER-REVIEW ACTIVITIES

- Surface and Coatings Technology
- Journal of Manufacturing Processes
- Additive Manufacturing
- Journal of Manufacturing Science and Engineering
- Optics & Laser Technology
- Optics & Lasers in Engineering
- Manufacturing Letters
- Metallurgical and Materials Transactions A
- Material Science & Engineering A
- Journal of Alloys & Compounds
- Applied Thermal Engineering
- Lasers in Manufacturing and Material Processing
- Ceramic International
- Progress in Additive Manufacturing
- Applied Surface Science Advances
- International Journal of Material Forming
- Advances in Materials and Processing Technologies
- Material Characterization
- Transactions of the Indian Institute of Metals
- Scientific Reports
- Sadhana
- Journal of Thermal Spray Technology

RESEARCH PROJECTS

Sl. No	PI/Co-PI	Project Title	Sponsoring Agency	Period	Cost (In lakhs)
1	PI	Simulation capabilities for additive manufacturing processes	DRDO	2024 - 2027	316.36

2	PI	Real-time monitoring and in-situ defect control in laser and resistance spot welding of Haynes 25 alloy for impingement tube of aero engine	ARDB	2024 - 2027	50.5
3	PI	Processes Development and Standardization for stress measurement on critical sections of turbine blades	Azad Engineering Ltd	2024 - 2027	9.34
4	Co-PI	3D printing of Conical Shape Charge Liner and lattice structure: Feasibility, Consistency and production scaling	DRDO	2024 - 2027	1676.34
5	PI	Printing of Ti6Al4V bottom Clamp using TIG based WAAM process	Bharath heavy electricals limited	2024-2024	9.91
6	PI	Services towards residual stress measurement in rotor blades	GTRE	2024 - 2024	1.9
7	PI	Development of hybrid friction stir welding process and hardfaced tools for onsite joining of high strength materials through assistance of high-power laser and additive manufacturing route	DST	2023 - 2025	43.06
8	PI	Laser cladding of functionally graded ceramic coating for high temperature and wear applications	DRDO	2021 - 2025	110.91
9	PI	Mechanical property evaluation & Metallurgical analysis of Laser Welded joints of E16NCD13 grade steel for Aero Engine Gear Application	GTRE	2021 - 2025	9.72
10	Co-PI	Large Area Additive Manufacturing (LAAM): Design and Development of Powder based Directed energy deposition system for direct fabrication of rocket components	DRDO	2021 - 2025	912.5
11	PI	Additive Manufacturing of Flexible push fits by Stereolithography process: Process optimization and product realization	Sandvik Rock Tools	2023 - 2024	2.29
12	Co-PI	Ultrasonic assisted laser additive manufacturing of nickel based super alloys and its online temperature monitoring to control the directionality in grain growth, anisotropy in mechanical properties and elemental segregation; and enhancement of the component life by laser shock peening	SERB	2020 - 2024	29.8
13	PI	Investigating the effect of shot peening process parameters and material heat treatment condition on the compressive stress induced along the depth of EN 36	Sandvik Rock Tools	2022 - 2023	24.15
14	PI	Real time monitoring and control of cooling rates in laser based additive manufacturing for ensuring microstructural homogeneity, controlling and dimensional accuracy, micro-cracks and delamination and investing their effects on the mechanical properties of fabricated component	SERB	2019 - 2021	27.94

PUBLICATION

PATENTS

S. No	Patent Title	Name of Applicant(s)	Patent No.	Award Date	Agency/ Country	Status
1	Real-time monitoring and control of thermal history using multiple pyrometers for laser material processing including additive	Ashish Kumar Nath, Muvvala Gopinath, Amal M Nair, Sagar Sarkar	559411	31/01/2025	India	Granted

	manufacturing processes					
2	Direct Additive Laser Welding of Dissimilar Materials	Angshuman Chattopadhyay, Muvvala Gopinath, Sagar Sarkar, Abhijit Sadhu, Vikranth Racherla, Ashish Kumar Nath	541488	11/06/2024	India	Granted
3	Rapid differential cooling for crack mitigation in laser welding of dissimilar metals of different thermal expansion coefficients	Angshuman Chattopadhyay, Muvvala Gopinath, Vikranth Racherla, Ashish Kumar Nath	491296	28/12/2023	India	Granted
4	Water Assisted Underwater Direct Laser Welding Technique and Processing Set-Up	Suvradip Mullick, Muvvala Gopinath, Parag Malhari Sakate, Ashish Kumar Nath	468267	10/11/2023	India	Granted

JOURNALS

S.No.	Title of the Publication	Name of the Journal	Authors list	Year of Publication	Volume, Issue, Pages	Impact Factor	Quartile
After Joining IIT Hyderabad							
1	A study on laser cladding of Stellite 6 by conventional and high-speed laser cladding process	Surface & Coatings Technology	Dharavathu, Raja, Kishora Shetty, and Gopinath Muvvala	2025	515, 132623	6.1	Q1
2	Laser–Material Interaction and Its Significance in the Laser-Directed Energy Deposition Process	Journal of Manufacturing Science and Engineering	Arkajyoti Jha, Ramji M, Gopinath Muvvala	2025	147, 11, 111001	2.9	Q1
3	Modelling and monitoring of scaling effects in multi-scan laser forming	Optics & Laser Technology	Bikram K Khandai, Muvvala Gopinath	2025	186, 112712	5	Q1
4	A study on laser welding of Inconel 718 and evolution of strain field using digital image correlation to estimate the localized properties	Optics & Laser Technology	Swapnil A Patil, Arkajyoti Jha, P Sankar Ganesh, Vijay Uttamrao Petley, S Rajeswari, SM Shariff, M Ramji, Gopinath Muvvala	2025	184, 112472	5	Q1
5	A Combined Approach of Melt Pool Ultrasonic Vibration and Interpass Laser Remelting for Achieving Superior Mechanical Properties by Controlling Microstructure and Phases in L-DED of Inconel 625	Journal of Manufacturing Science and Engineering	Prabhat Kumar, Shubham Rao, Srikant Gollapudi, Gaurav Bartarya, Gopinath Muvvala, Suvradip Mullick	2025	147, 3, 031013	2.9	Q1
6	A study on the deformation heterogeneity in laser-welded EN16NCD13 grade steel analyzed using digital image correlation	Welding in the World : The International Journal of Materials Joining	Bikram K Khandai, Yash R Tandel, Arkajyoti Jha, Vijay Uttamrao Petley, S Rajeswari, M Ramji, Muvvala Gopinath	2025	1-18	2.5	Q2
7	Assessing the efficacy of laser modulation in controlling the anisotropy through monitoring of molten pool state using pyrometer in laser-directed energy deposition	The International Journal of Advanced Manufacturing Technology	Arkajyoti Jha, M Ramji, Muvvala Gopinath	2025	136, 5, 2559-2582	3.1	Q1
8	An investigation on residual stress distribution in cold formed steel channel section	Journal of Structural Engineering	Alok Panditaray; Bishal Naik; Mahendrakumar Mathialagu Madhavan; Gopinath Muvvala	2025	Accepted	4.2	Q1

8	Improvement in mechanical and corrosion properties of laser and electron beam welded AISI 304 stainless steel joints by laser shock peening	Lasers in Manufacturing and Materials Processing	Angshuman Chattopadhyay, Sagar Sarkar, Gopinath Muvvala, Harikrishnan Ramadas, Vikranth Racherla, Ashish Kumar Nath	2024	11, 4, 946-983	2.43	Q2
9	Understanding in-process responses in multi-layer friction stir additive manufacturing: temperature, viscosity, tool torque, and mechanical properties	Journal of Materials Processing Technology	Ram Rapaka, Harish Ladi, Dharavathu Raja, Gopinath Muvvala, Tuhin Mukherjee, Buchibabu Vicharapu	2024	330, 118491	7.5	Q1
10	Continuous and Pulse TIG Arc Treatment for Surface Hardening of WAAM-MIG Parts	Transactions of the Indian Institute of Metals	Anas Ullah Khan, Satyajit Chatterjee, Muvvala Gopinath, Yuvraj K Madhukar	2024	77, 8, 2163-2172	1.6	Q2
11	Real-time monitoring of temperature gradients and bending mechanism in multi-scan laser forming process	Journal of Manufacturing Processes	Bikram K Khandai, Shivam Shukla, Gopinath Muvvala	2024	119, 975-986	6.8	Q1
12	Effect of ultrasonic vibration on microstructural evolution, clad defects, and surface properties in laser direct energy deposition of Inconel 625	Journal of Laser Applications	Biplab Kumar Dash, Shrey Bhatnagar, Hari Srinivasa Rao Magham, Shubham Rao, Gopinath Muvvala, Suvradip Mullick	2024	36, 2	2.1	Q2
13	Investigating the effect of laser modulation and input energy on bending mechanism and bending angle in the multi-pass laser forming process through real-time monitoring	Optics & Laser Technology	Bikram K Khandai, Shivam Shukla, Gopinath Muvvala	2024	169, 110100	5	Q1
14	The effect of different laser head angles and shielding gas supply systems to maximize the depth of penetration by minimizing the plasma shielding effect in fiber laser welding of 3-mm thick NiTiInol sheet	Optik	Susmita Datta, Mohammad Shahid Raza, Gopinath Muvvala, Partha Saha, Dilip Kumar Pratihari	2023	283, 170903	8.3 (Cite Score)	Q1
15	Evaluation of microstructure and thermal history for TiC/Inconel 625 MMC deposition through pre-placed laser cladding method with and without the application of ultrasonic vibration	CIRP Journal of Manufacturing Science and Technology	Shrey Bhatnagar, Hari Srinivasarao Magham, Suvradip Mullick, Muvvala Gopinath	2023	41, 453-464	5.4	Q1
16	A study on developing process-structure-property relationship with molten pool thermal history during laser surface remelting of Inconel 718	Optics & Laser Technology	Arkajyoti Jha, Shivam Shukla, Amit Choudhary, Ramji Manoharan, Gopinath Muvvala	2023	157, 108732	5	Q1
17	A study on anisotropy in wire arc additively manufactured Inconel 625 multi-layered wall and its correlation with molten pool thermal history	Materials Science and Engineering: A	Sagar Pawar Dadasaheb, Srinath Ellaswamy Gudur, Vishwanath Nagallapati, Amit Choudhary, Arun Torris, Gopinath Muvvala	2022	840, 142865	7	Q1
18	Angshuman Chattopadhyay, Gopinath Muvvala, Sagar Sarkar, Vikranth Racherla, Ashish Kumar Nath	Journal of Materials Processing Technology	Angshuman Chattopadhyay, Gopinath Muvvala, Sagar Sarkar, Vikranth Racherla, Ashish Kumar Nath	2022	300, 117403	7.5	Q1

19	Microstructure, mechanical, and corrosion properties of electron beam-welded commercially pure titanium after laser shock peening	The International Journal of Advanced Manufacturing Technology	Angshuman Chattopadhyay, Gopinath Muvvala, Sagar Sarkar, Vikranth Racherla, Ashish Kumar Nath	2022	118, 1, 343-364	3.1	Q1
20	Investigating the effect of process parameters on weld pool thermal history and mechanical properties of laser welded Inconel 625 and Duplex stainless steel 2205 dissimilar welds	Optik	Gulshad Nawaz Ahmad, Mohammad Shahid Raza, NK Singh, Gopinath Muvvala	2022	248, 168134	8.3 (Cite Score)	Q1
21	A lumped parametric analytical model for predicting molten pool temperature and clad geometry in pre-placed powder laser cladding	Optik	Shrey Bhatnagar, Suvradip Mullick, #Muvvala Gopinath	2021	247, 168015	8.3 (Cite Score)	Q1
22	A study on the effect of molten pool thermal history and gaseous environment in laser surface alloying of AISI 1020 steel with TiN	Optik	Vijayababu Vommi, Amit Choudhary, Gopinath Muvvala	2021	245, 167758	8.3 (Cite Score)	Q1
23	An investigation on physical phenomena of water-jet assisted underwater wet laser welding technique under continuous and pulsed mode operation	Optik	Parag Malhari Sakate, Suvradip Mullick, Muvvala Gopinath	2021	242, 167272	8.3 (Cite Score)	Q1
24	High-temperature abrasive wear characteristics of H13 steel modified by laser remelting and clad with Stellite 6 and Stellite 6/30% WC	Surface and Coatings Technology	Debapriya Patra Karmakar, Gopinath Muvvala, Ashish Kumar Nath	2021	127498	6.1	Q1
25	Monitoring and analysis of melt-assisted deformation behavior of 304L stainless steel during multipass laser forming process using IR pyrometer and laser-based displacement sensor	Optics & Laser Technology	Mohammad Shahid Raza, Susmita Datta, Muvvala Gopinath, Partha Saha	2021	135, 106718	5	Q1
26	Online monitoring of thermo cycles during laser remelting of flame sprayed chromia coating in pulsed mode and coating properties	Optik	Biswajit Das, Muvvala Gopinath, Ashish Kumar Nath, PP Bandyopadhyay	2021	227, 166030	8.3 (Cite Score)	Q1
27	Laser surface polishing of NiCrSiBC – 60WC ceramic-metal matrix composite deposited by laser directed energy deposition process	Surface and Coatings Technology	Choudhary, Amit, Abhijit Sadhu, Sagar Sarkar, Ashish Kumar Nath, and Gopinath Muvvala	2020	404, 126480	6.1	Q1
29	Real-time detection of cooling rate using pyrometers in tandem in laser material processing and directed energy deposition	Materials Letters	Amal M Nair, Gopinath Muvvala, Sagar Sarkar, Ashish Kumar Nath	2020	277, 128330	2.7	Q2
30	Development of process maps based on molten pool thermal history during laser cladding of Inconel 718/TiC metal matrix composite coatings	Surface and Coatings Technology	Gopinath Muvvala, Suvradip Mullick, Ashish Kumar Nath	2020	399, 126100	6.1	Q1
31	Effect of laser shock peening on microstructural, mechanical and corrosion properties of laser beam	Optics & Laser Technology	Angshuman Chattopadhyay, Gopinath Muvvala, Sagar	2020	133, 106527	5	Q1

	welded commercially pure titanium		Sarkar, Vikranth Racherla, Ashish Kumar Nath				
32	Performance of additive manufactured Stellite 6 tools in friction stir processing of CuCrZr sheet	Optics & Laser Technology	Abhijit Sadhu, Debapriya Patra Karmakar, Omkar Mypati, Gopinath Muvvala, Surjya K Pal, Ashish Kumar Nath	2020	128, 106241	5	Q1
33	A study on the influence of substrate pre-heating on mitigation of cracks in direct metal laser deposition of NiCrSiBC-60% WC ceramic coating on Inconel 718	Surface and Coatings Technology	Abhijit Sadhu, Amit Choudhary, Sagar Sarkar, Amal M Nair, Pravanjan Nayak, Sagar Dadasahed Pawar, Gopinath Muvvala, Surjya K Pal, Ashish Kumar Nath	2020	389, 125646	6.1	Q1
34	Effect of scan strategy and heat input on the shear strength of laser clad Stellite 21 layers on AISI H13 tool steel in as-deposited and heat treated conditions	Surface and Coatings Technology	Debapriya Patra Karmakar, Gopinath Muvvala, Ashish Kumar Nath	2020	384, 125331	6.1	Q1
35	A study on in-situ synthesis of TiCN metal matrix composite coating on Ti-6Al-4V by laser surface alloying process	Journal of Alloys and Compounds	Amal M. Nair, Muvvala Gopinath, Ashish Kumar Nath	2019	810, 151901	6.3	Q1
Before Joining IIT Hyderabad							
36	Role of molten pool thermo cycle in laser surface alloying of AISI 1020 steel with in-situ synthesized TiN	Surface and Coatings Technology	Muvvala Gopinath, Pavan Thota, Ashish Kumar	2019	362, 150-166	6.1	Q1
37	Effect of tempering on laser remelted AISI H13 tool steel	Surface and Coatings Technology	Debapriya Patra Karmakar, Gopinath Muvvala, Ashish Kumar Nath	2019	361, 136 - 149	6.1	Q1
38	Effect of cooling rate on residual stress and mechanical properties of laser remelted ceramic coating	Journal of the European Ceramic Society	Biswajit Das, Muvvala Gopinath, Ashish Kumar Nath, PP Bandyopadhyay	2018	38, 11, 3932 - 3944	6.2	Q1
39	In-process detection of microstructural changes in laser cladding of in-situ Inconel 718/TiC metal matrix composite coating	Journal of Alloys and Compounds	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2018	740, 545-558	6.3	Q1
40	Monitoring and assessment of tungsten carbide wettability in laser clad metal matrix composite coating using an IR pyrometer	Journal of Alloys and Compounds	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2017	714, 514-521	6.3	Q1
41	Online assessment of TiC decomposition in laser cladding of metal matrix composite coating	Materials & Design	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2017	121, 310-320	7.9	Q1
42	Online monitoring of thermo-cycles and its correlation with microstructure in laser cladding of nickel based super alloy	Optics and Lasers in Engineering	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2017	88, 139-152	5	Q1
43	Analysis of temperature and surface hardening of low carbon thin steel sheets using Yb-fiber laser	Surface and Coatings Technology	Sagar Sarkar, Muvvala Gopinath, Shitanshu S Chakraborty, Badirujaman Syed, Ashish K Nath	2016	302, 344-358	6.1	Q1

BOOK CHAPTER

S.No	Title	Author's Name	Publisher	Year of Publication
1	Laser Cladding of Ni-Based Superalloys https://doi.org/10.1007/978-3-030-53195-9_11 .	Muvvala G., Sarkar S., Nath A.K.	Springer	2021
2	LASER based manufacturing as a green manufacturing process https://doi.org/10.1201/9781315163147 .	Ashish Kumar Nath, Shitanshu Chakraborty, Sagar Sarkar, Gopinath Muvvala, Suhradip Mullick, Yuvraj K Madhukar, Debapriya Patra Karmakar	CRC Press	2019

CONFERENCE PAPERS

S. No	Title of paper	Name of the conference	Authors list	Year of Conference
After Joining IIT Hyderabad				
1	Quantitative analysis of localized strength variation in friction stir additive manufactured Aluminum alloy: Effects of Re-stirring on mechanical properties	4th Asia-Pacific International Conference on Additive Manufacturing (APICAM), RMIT University, Australia	Dharavathu Raja, Muvvala Gopinath	2025
2	Laser Powder Bed Fusion of Pure Copper Using Green Laser	2nd International Conference on Future of AM, Singapore	Bikram K. Khandai, Arkajyoti Jha, and Muvvala Gopinath	2025
3	Machine Learning-Based Control System Simulation for In-Situ Monitoring of Additive Manufacturing Process	13th International Conference on Precision, Meso, Micro and Nano Engineering, Copen	Aditya Manoj Madgulkar, Muvvala Gopinath	2024
4	AI-Driven Real-Time Cooling Rate Control for Optimized Additive Manufacturing Process and Sustainable Production	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Aditya Manoj Madgulkar, Muvvala Gopinath	2024
5	Investigating the bending behaviour of SS304 in a multi-pass laser forming process through real-time monitoring and finite element model	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Bikram K. Khandai, Arkajyoti Jha, and Muvvala Gopinath	2024
6	A study on deformation inhomogeneity of laser welded low carbon steel through real-time monitoring of molten-pool and digital image correlation	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Bikram K. Khandai, Arkajyoti Jha, and Muvvala Gopinath	2024
7	A Study on Influence of Shielding Gas on Microstructure and Mechanical Properties of Additively Manufactured Inconel 718	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Ashish Parihar, Arkajyoti Jha, Gopinath Muvvala	2024
8	High-speed additive manufacturing of stellite-6 tools for Friction stir welding of high-strength materials	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Prasannakumar Biradar, Muvvala Gopinath	2024
9	Mitigation of tunneling defects and analysis of mechanical behavior in double-sided friction stir welded Al 2014 plates	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Dharavathu Raja, Muvvala Gopinath	2024
10	A study on Online Monitoring of Modulated Mode Laser for reducing Anisotropy in Laser-Directed Energy Deposition	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Arkajyoti Jha, Bikram K. Khandai, Muvvala Gopinath, and M. Ramji	2024
11	A study on the effect of Heat Treatment conditions on the mechanical properties of WAAM-repaired Inconel 718	Processing of Advanced Materials and Fabrication of Products (PAMFP)	Arkajyoti Jha, Bikram K. Khandai, Muvvala Gopinath, and M. Ramji	2024
12	Laser surface polishing of additively manufactured components through laser beam shaping	International Conference on Materials Processing using Lasers and Surface Engineering (IMPULSE 2023), IIT Madras	Arkajyoti Jha, Vignan Babu Dyavanapelly, Bikram K Khandai, Muvvala Gopinath	2023

13	A Study on the Effect of Modulated Mode Laser on Anisotropy in Laser-Directed Energy Deposition	International Conference on Materials Processing using Lasers and Surface Engineering (IMPULSE 2023), IIT Madras	Arkajyoti Jha, Dharavathu Raja, Bikram K Khandai, Muvvala Gopinath	2023
14	Strategies to Control Porosity in Laser Directed Energy Deposition	International Conference on Materials Processing using Lasers and Surface Engineering (IMPULSE 2023), IIT Madras	Arkajyoti Jha, Jagdish Chandra Pandey, Bikram K Khandai, Muvvala Gopinath	2023
15	Controlling waviness in additive manufacturing of thin walls by laser directed energy deposition process	All India Manufacturing Technology, Design and Research Conference (AIMTDR), India	Shivam Shukla, Srinath Gudur, J John Rozario Jegaraj, P Mastanaiah, Muvvala Gopinath, Suryakumar Simhambhatla	2021
16	A study on the effect of substrate heating and cooling on bead geometry in wire arc additive manufacturing and its correlation with cooling rate	Material Today Proceedings	Srinath Gudur, Vishwanath Nagallapati, Sagar Pawar, Gopinath Muvvala, Suryakumar Simhambhatla	2021
17	A study on the effect of cooling rate on evolution of microstructure in laser surface remelting of Inconel 718	ICRAMERD	Arkajyoti Jha, Amit Choudhary, Batta Renuka, Muvvala Gopinath	2020
18	A study on mitigation of cracks in multilayered direct metal laser deposition of NiCrBSiC-60%WC on Inconel 718	International Conference on Precision, Meso, Micro and Nano Engineering (COPEN), IIT Indore	Abhijit Sadhu, Amal M Nair, Satya Srinivas Ravada, Amit Choudhary, Muvvala Gopinath, Surjya K Pal, Ashish Kumar Nath	2019
19	Laser welding of Ti and AISI 304 with Ni interlayer	International Conference on Precision, Meso, Micro and Nano Engineering (COPEN), IIT Indore	Chattopadhyay A, Gopinath M, Racherla V, Nath A K	2019
20	Laser Surface Polishing of NiCrSiBC-60WC Ceramic-Metal Composite Deposited by Direct Metal Laser Deposition	Visual-JW 2019 & WSE 2019, JWRI, Osaka University, Japan	Abhijit Sadhu, Amit Choudhary, Sagar Sarkar, Pawar Sagar Dadasaheb, Masetty Krishna Chaitanya, Muvvala Gopinath, Surjya K Pal, Ashish Kumar Nath	2019
21	Assessment of Microstructure and Mechanical Properties in Laser Cladding, Surface Polishing and Welding Through Online Monitoring	Visual-JW 2019 & WSE 2019, JWRI, Osaka University, Japan	Muvvala Gopinath, Ashish Kumar Nath	2019
22	Effect of laser shock peening on corrosion properties of laser additive manufactured IN718 superalloy	Proceedings of LAMP2019 – The 8th International Congress on Laser Advanced Materials Processing	Sagar Sarkar, Harikrishnan R, Muvvala Gopinath, Ashish Kumar Nath	2019
23	Effect of process parameters on geometrical aspects in direct metal laser deposition of Ni5Mo5Al hardface coating	IOP Conf. Series: Materials Science and Engineering	Vikash Kumar, Ankit Shrivastava, Debapriya Patra Karmakar, Shitanshu Shekhar Chakraborty, Himadri Roy, Muvvala Gopinath, Prakash Kumar, Sumanta Mukherjee	2019
24	Assessment Of Microstructure And Mechanical Properties In Laser Surface Remelting And Cladding Through Online Monitoring Of Thermal History	ASME 14th International Manufacturing Science and Engineering Conference	Ashish Kumar Nath, Muvvala Gopinath	2019
Before Joining IIT Hyderabad				
25	A study on the influence of shielding gas on TiN decomposition in laser surface alloying	National Conference on Advanced Materials, Manufacturing and Metrology	Muvvala Gopinath, G Sai Krishna, Ashish Kumar Nath	2018

26	Monitoring of Molten Pool Thermal History and Its Significance in Laser Cladding of Inconel718 Based Composite Coatings	National Laser Symposium (NLS-27)	Muvvala Gopinath	2018
27	Study of molten pool thermal history during laser welding of titanium and AISI 304	National Laser Symposium (NLS-27)	A. Chattopadhyay, M. Gopinath, V. Racherla, A. K. Nath	2018
28	A study on in-situ synthesis of TiCN metal matrix composite coating on Ti-6Al-4V by laser alloying process	National Laser Symposium (NLS-27)	Amal M Nair, Muvvala Gopinath, Pravanjan Nayak, Ashish Kumar Nath	2018
29	Effect of overlapping tracks on the microstructure and thermal history of predecessor track in multi-track laser surface remelting	National Laser Symposium (NLS-27)	Debapriya Patra Karmakar, Muvvala Gopinath, Ashish Kumar Nath	2018
30	A study on laser surface alloying of AISI 1020 mild steel with in-situ synthesized TiN	Advances in Materials & Processing Technologies Conference	Muvvala Gopinath, Pavan Thota, Ashish Kumar Nath	2018
31	Effect of laser cladding parameters on clad-track uniformity	National Conference on Advanced Materials, Manufacturing and Metrology (NCAMMM)	Debapriya Patra Karmakar, Gopinath Muvvala, Shams Perwez, Ashish Kumar Nath	2018
32	Development of Friction Stir Welding Tools using Direct Metal Laser Deposition Technique	Advances in Materials & Processing Technologies Conference	Abhijit Sadhu, Debapriya Patra Karmakar, Gopinath Muvvala, Omkar Mypati, Surjya Kanta Pal, Ashish Kumar Nath	2018
33	A study on laser welding of titanium and stainless steel	MSEC, College station, TX, USA	Angshuman Chattopadhyay, Muvvala Gopinath, Vikranth Racherla, Ashish Kumar Nath	2018
34	Monitoring of Molten Pool Thermal History and its Significance in Laser Cladding Process	ASME Proceedings-Additive Manufacturing, MSEC2017-2657	Gopinath Muvvala, Debapriya Patra Karmakar, Ashish Kumar Nath	2017
35	Investigation of Wear and Corrosion Characteristics of Stellite-6 and Stellite-21 Layers Deposited by Co-Axial Laser Cladding	ASME Proceedings-Additive Manufacturing, MSEC2017-2841	Debapriya Patra Karmakar, Muvvala Gopinath, Soham Harmalkar, Ashish Kumar Nath	2017
36	Effect of Tempering Temperature on Hardness and Microstructure of Laser Surface Remelted AISI H13 Tool Steel	ASME Proceedings-Additive Manufacturing, MSEC2017-3014	Debapriya Patra Karmakar, Muvvala Gopinath, Ashish Kumar Nath	2017
37	A Study towards Online Detection of Evolution of Microstructure in In-Situ and Ex-Situ Processing of TiC Metal Matrix Composite Coating by Laser Cladding Technique	National Conference on Advanced Functional Materials Processing & Manufacturing (NCAFMPM)	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2017
38	Monitoring of Clad-Track Geometry using Temperature Signals for Co-Axial Laser Cladding	National Conference on Advanced Functional Materials Processing & Manufacturing (NCAFMPM)	Debapriya Patra Karmakar, Muvvala Gopinath, Ashish Kumar Nath	2017
39	Striation-free cutting of mild steel and stainless steel by Yb-Fiber laser	National Laser Symposium (NLS-25)	S Mullick, A Priyadarshini, M Gopinath, A K Nath	2016
40	Performance comparison of surface modification by surface remelting and stellite-6 cladding using fiber laser for H13 Tool steel substrate	National Laser Symposium (NLS-25)	Debapriya Patra Karmakar, Muvvala Gopinath, Ashish Kumar Nath	2016
41	A study towards in-situ monitoring of TiC particle decomposition in laser cladding of metal matrix composite coating	National Laser Symposium (NLS-25)	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2016
42	A study on geometrical aspects, microstructures and its correlation with surface temperatures in preplaced powder cladding of nickel based super alloy	MATADOR, Taiwan	Gopinath Muvvala, D Chandran, B Malaya Prasad, A K Nath	2015

43	Effect of Pulse Width on Clad Track Geometry and Surface Roughness in Pre-placed Powder Laser Cladding of Inconel 718	Lasers and Plasma Application in Materials Science	Gopinath Muvvala, B Malaya Prasad, A K Nath	2015
44	Development and characterization of a coaxial laser cladding nozzle	National Laser Symposium (NLS-24)	Muvvala Gopinath, Debapriya Patra Karmakar, Ashish Kumar Nath	2015
45	Parameter optimization for laser cutting and drilling of carbon fiber reinforced plastic sheets with fiber laser	National Laser Symposium (NLS-22)	Vivek Sinha, Harshad Padmawar, Gopinath Muvvala, Ashish K Nath	2014
46	Analytical and Finite Element Modeling of Pre-Placed Powder Laser Cladding	Kiran: A Bulletin of the Indian Laser Association	M. Gopinath, S.S. Chakraborty, D. Chandran, A.K. Nath	2014
47	Surface Hardening of Ti-6Al-4V During Turning Via Pyrolytic Carburization	International Conference on Engineering Materials and Processes (ICEMAP)	Sakhamuri V Kishore, S Oliver Nesa Raj, Gopinath Muvvala	2013

LIST OF COURSES TAUGHT

S No	Acad Year	Course Code	Course Name	Credits	Score
1	2024-25	ME5200	Additive Manufacturing	1.5	4.60
2	2024-25	ME5250	Design for Manufacturing and Assembly	1.5	4.59
3	2024-25	ME3220	Industrial Engineering and Operations Research	3	4.47
4	2024-25	ME5421	Finite Element Methods Lab	1	4.14
5	2024-25	ME5909	Additive Manufacturing	2	4.47
6	2024-25	AM5030	Materials for Additive Manufacturing	2	4.56
7	2023-24	ME5909	Additive Manufacturing	2	4.00
8	2023-24	ME3220	Industrial Engineering and Operations Research	3	4.22
9	2023-24	AM5030	Materials for Additive Manufacturing	2	4.72
10	2023-24	ME5421	Finite Element Methods Lab	1	4.13
11	2023-24	AM5100	Industrial Lectures	1	4.84
12	2023-24	ME5250	Design for Manufacturing and Assembly	1.5	4.60
13	2022-23	AM5100	Industrial Lectures	1	4.49
14	2022-23	ME5200	Additive Manufacturing	1.5	4.59
15	2022-23	AM5100	Industrial Lectures	1	4.49
16	2022-23	ME5421	Finite Element Methods Lab	1	4.53
17	2022-23	ME5200	Additive Manufacturing	1.5	4.59
18	2022-23	ME3050	Computer Integrated Manufacturing	1.5	3.77
19	2022-23	ME5909	Additive Manufacturing	2	4.07
20	2022-23	ME3010	Manufacturing Science - II	2	4.09
21	2022-23	ME3465	Manufacturing Lab	1	4.14
22	2022-23	ME5210	CAD/CAM	3	4.60
23	2021-22	ME5200	Additive Manufacturing	1.5	4.17
24	2021-22	ME5421	Finite Element Methods Lab	1	4.23
25	2021-22	ME5720	Advanced Material Joining Processes	1.5	4.40
26	2021-22	ME5431	Integrated Design & Manufacturing Lab	2	4.47

27	2021-22	ME5200	Additive Manufacturing	1.5	4.17
28	2021-22	ME5421	Finite Element Methods Lab	1	4.23
29	2021-22	ME5720	Advanced Material Joining Processes	1.5	4.40
30	2021-22	ME5431	Integrated Design & Manufacturing Lab	2	4.47
31	2021-22	ME3010	Manufacturing Science - II	2	3.79
32	2021-22	ME3050	Computer Integrated Manufacturing	1.5	3.96
33	2021-22	ME5909	Additive Manufacturing	2	4.44
34	2021-22	ME5210	CAD/CAM	3	4.64
35	2021-22	AM5041	Additive Manufacturing Processes Lab	1	4.73
36	2020-21	ME5720	Advanced Material Joining Processes	1.5	4.57
37	2020-21	ME3465	Manufacturing Lab	1	3.77
38	2020-21	ME5200	Additive Manufacturing	1.5	4.41
39	2020-21	ME5720	Advanced Material Joining Processes	1.5	4.57
40	2020-21	ME4325	Project-2	3	3.84
41	2020-21	ME5080	Scaling Laws & Multi-scale Manufacture	1.5	4.25
42	2020-21	ME3010	Manufacturing Science - II	2	4.27
43	2019-20	ME3465	Manufacturing Lab	1	3.63
44	2019-20	ME5720	Advanced Material Joining Processes	1.5	4.18
45	2019-20	ME5431	Integrated Design & Manufacturing Lab	2	4.51
46	2019-20	ME3010	Manufacturing Science - II	2	3.83